

Polina Kliuchnikova

ML Researcher | RLHF, Game Theory & Applied Mathematics

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Research: istina.msu.ru/workers/251795028/ | Earlier publications under Polina S. Tanasevich

RESEARCH PROFILE

PhD student in AI & Machine Learning at ITMO University and Staff ML Engineer at Huawei. Research spans game-theoretic models of human–AI interaction, reinforcement learning from human feedback (RLHF) and statistical identification of human preferences. Combine mathematical modelling with hands-on LLM post-training. Interested in scalable oversight, particularly how evaluator incentives and unreliable feedback affect supervision.

SELECTED RESEARCH

Game-theoretic models of human oversight

- Modelled evaluator–LLM incentives as a 5×5 strategic-form game; computed mixed-strategy Nash equilibria by support enumeration and analysed best responses and replicator dynamics.
- Studied delegation versus verification in a 2×2 human–AI role game using Nash and logit quantal response equilibria (QRE). Analysed sensitivity across 2,772 model variants and investigated structural non-identifiability of payoff weights using algebra and parameter-recovery simulations.

Auditing the human feedback used to train language models

- Co-developed a bias-auditing framework for RLHF code evaluation: per-rater reliability estimation, permutation tests with false discovery rate control, and a Markov-chain Bayesian quality filter.
- Studied 305 sessions from 50 raters on 267 CoNaLa code pairs; validated the quality estimator with 20,000 Monte Carlo replications. Calibration changed 3.7% of pairwise quality verdicts, with external checks against execution- and reference-based signals. Manuscript under review.

Identifiability of human preferences and reliance on AI

- Coauthored a formal analysis of when predictive regression coefficients fail to identify feasible preference trade-offs. Developed profile-likelihood and support-based diagnostics, evaluated through simulations and 1,720 code-adoption judgements from 10 evaluators. Manuscript under review.

Cognitively informed reinforcement learning

- Coauthored work integrating a cognitive critic into reinforcement learning from AI feedback (RLAIF) for code generation; compared supervised fine-tuning, contrastive learning and reward models, including Markov-chain reward aggregation. Article in press.

PROFESSIONAL EXPERIENCE

Huawei | Staff ML Engineer | Sep 2025–Present

- Developed RLHF post-training pipelines with PPO, reward modelling, synthetic data generation and training-data quality controls.
- Fine-tuned and post-trained LLMs using multi-GPU DeepSpeed and mixed precision; deployed inference with vLLM. Built evaluation workflows and benchmarks for generated outputs.

Sberbank | Tech Lead, Machine Learning | Feb 2024–Aug 2025

- Led ML engineering work across LLM-based geo-analytics and automated log analysis, taking models from research to deployment; optimised GPU inference using TensorRT and reduced precision.

SELECTED PUBLICATIONS & MANUSCRIPTS

Published work, in-press work and submissions are distinguished below. Titles of Russian-language abstracts are translated into English.

In press

S. V. Kovalchuk, D. V. Fedrushkov, P. S. Kliuchnikova and A. T. S. Ireddy. Adaptation of Artificial Intelligence Agents to Individual Operators's Cognitive States Using Reinforcement Learning. In press.

Under review

S. V. Kovalchuk and P. S. Kliuchnikova. Annotator Bias Auditing for RLHF: Methodological Approach to Code-Evaluation Adjustment. Manuscript under review.

When People Don't Trade Off What We Think They Do: Support-Feasible Substitution on Bounded Adoption Surfaces, with a Case in AI-Generated Code. Coauthored manuscript under review.

Published

M. Ya. Yakovlev, P. S. Tanasevich, A. V. Vershinin and V. A. Levin. Numerical analysis of the effective thermal properties and the stability for NTE metamaterials using CAE fidesys. AIP Conference Proceedings 2509, 020210 (2022).
doi.org/10.1063/5.0084835

P. S. Tanasevich, M. Ya. Yakovlev and A. V. Vershinin. Solving the inverse problem of estimating effective properties of auxetic metamaterials using machine learning algorithms. XIII All-Russian Congress on Theoretical and Applied Mechanics, pp. 822–824 (2023). Conference abstract; in Russian.

EARLIER EXPERIENCE & MATHEMATICAL RESEARCH

VK / MY.GAMES | Senior Data Analyst | Oct 2021–Sep 2023

Developed lifetime-value and retention models; automated analytical data pipelines with Spark and Airflow and supported product experimentation.

CAE Fidesys | C++ Core Developer | Sep 2020–Aug 2021

Implemented computational mechanics algorithms for effective material properties. Coauthor of two registered software programs: RU 2020665248 and RU 2020667788 (2020).

Lomonosov Moscow State University | Researcher | Dec 2019–Aug 2021

Research in computational mechanics and numerical modelling of heterogeneous materials: finite-element analysis, thermoelasticity, effective properties and stability of metamaterials.

EDUCATION & ADDITIONAL TRAINING

Doctoral studies, AI & Machine Learning (in progress) | ITMO University

MSc, Applied Mathematics | Lomonosov Moscow State University | 2021–2023

BSc, Applied Mathematics | Lomonosov Moscow State University | 2017–2021

Large Language Model Agents MOOC | UC Berkeley RDI | Sep 2024–Jan 2025. Certificate of completion.

METHODS & TOOLS

Game theory: strategic-form games, Nash equilibria, support enumeration, quantal response equilibrium, best-response analysis, replicator dynamics, comparative statics.

Mathematics & statistics: numerical methods, finite elements, inverse problems, Bayesian updating, Markov chains, Monte Carlo simulation, permutation tests, profile likelihood, identifiability analysis.

ML & engineering: Python, C++, PyTorch, RLHF/RLAIF, PPO, SFT, reward modelling, evaluation design, DeepSpeed, vLLM, TensorRT, Docker, Linux, SQL, Spark, Airflow.